Network

Execution layer

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Arc's Reth-based execution layer processes transactions, maintains state, and validates results within the EVM environment.

Arc's execution layer is based on **Reth**, a Rust implementation of the Ethereum execution client. This layer maintains the ledger and blockchain state, executes transactions, and extends the EVM with Arc-specific modules for stablecoin-native finance.

As a developer, you don't interact directly with the execution layer internals, but understanding how it works helps you design apps that rely on Arc's predictable fees, privacy, and financial primitives.

Core responsibilities

The Execution layer has three primary jobs:

1. **Maintain the ledger and application state**
   - Tracks accounts, balances, and smart contracts.
   - Stores contract code and persistent state variables.
   - Records every transaction and resulting state change.
2. **Execute transactions**
   - Applies EVM logic for smart contracts and transfers.
   - Deducts gas costs using the Fee Manager.
   - Calls Arc modules (Privacy, Stablecoin Services) where applicable.
3. **Validate results**
   - Ensures transactions are valid before they can be finalized by consensus.
   - Rejects invalid transactions (for example, insufficient funds or failed contract logic).
   - Produces a state root hash, which consensus finalizes.

Arc-specific modules

Arc extends the base Ethereum execution model with the following modules:

- **Fee Manager:** Stabilizes fees by using USDC as the unit of account and smoothing fee changes.
- **Privacy Module:** Enables confidential transfers with encrypted amounts and selective disclosure through view keys.

  The Privacy Module is planned and not yet available on Arc.

- **Stablecoin Services:** Provide core stablecoin-native features such as cross-currency settlement, paymaster-style sponsored transactions, and multi-stablecoin gas payments.

  Stablecoin Services are planned and not yet available on Arc.

These modules are integrated into the execution pipeline, so you can rely on them without building custom extensions.

How Reth works

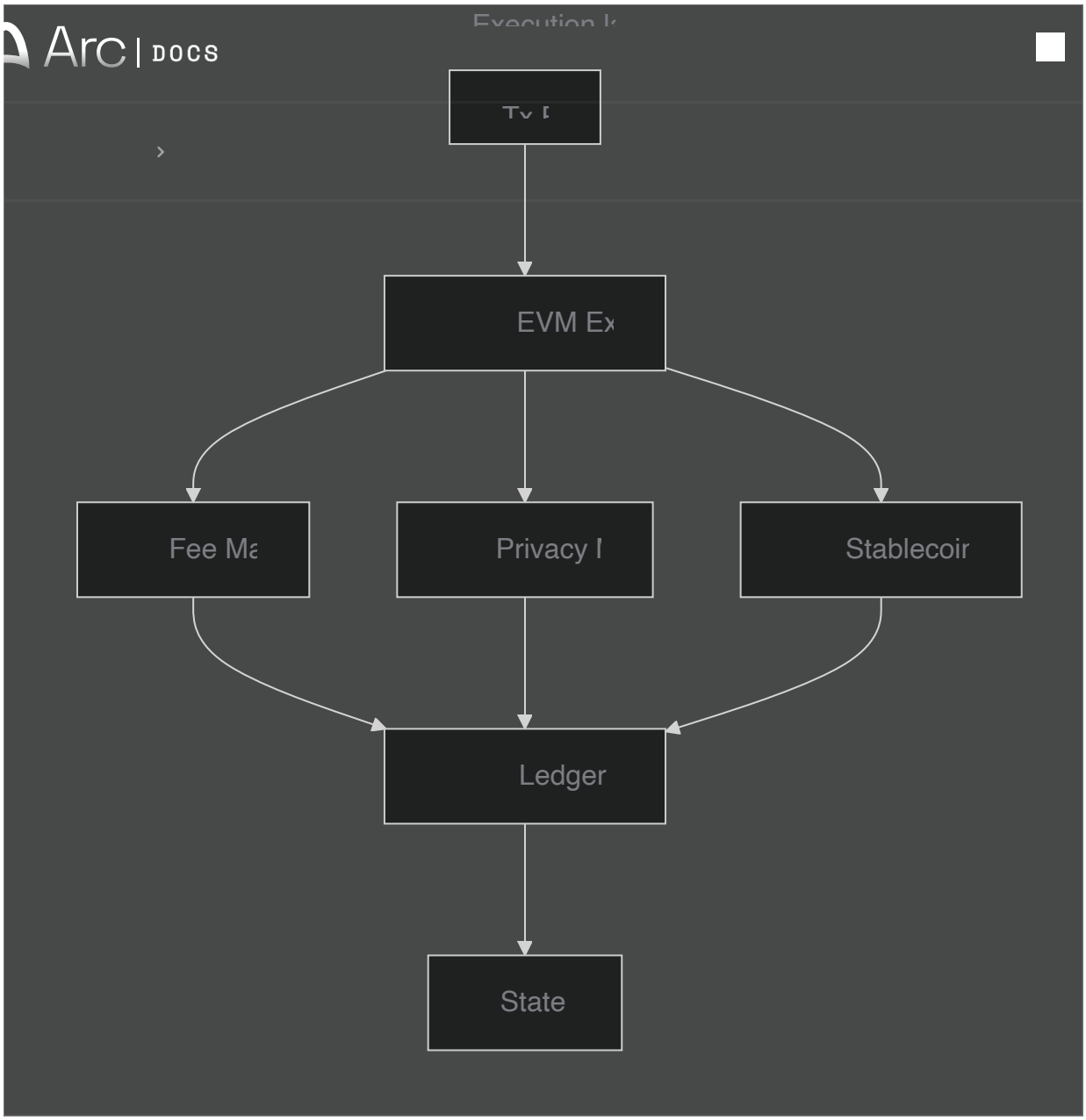
Reth, like other Ethereum execution clients, follows a structured pipeline:

1. **Transaction pool:** Holds pending transactions waiting to be included in a block.
2. **Block execution:** Applies transactions sequentially to the current state, updating balances, contract storage, and logs.
3. **Gas accounting:** Deducts gas fees through Arc's Fee Manager.
4. **Module calls:** Routes to Privacy or Stablecoin Services logic as required.
5. **State root:** Produces a Merkle root of the updated state.

Reth is written in Rust, which provides performance, safety, and modularity. Arc leverages this foundation to extend the execution layer with stablecoin-native features.

Execution diagram

The diagram below shows the flow of a transaction through the Execution layer.



Developer benefits

For developers, the Execution layer means:

- You build on a familiar EVM-compatible platform.
- Fees, privacy, and multi-stablecoin settlement are native features, not bolt-ons.
- Transactions are validated and applied consistently, following the order finalized by the consensus layer.
- The underlying ledger and state are managed efficiently by Reth, written in Rust for performance and reliability.

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